Problem: Find a value of the constant K for which the following function is continuous at $x=1$:

$$f(x) = \begin{cases} 7x-2 & , x \le 1 \\ Kx^2 & , x > 1 \end{cases}$$

Sol^n: [First we show the function is defined at $x=1$.

$$f(1) = 7.1-2 = 5 \text{] (No need)}$$

Now, $\text{L.H.L.} = \lim_{x \to 1^-} f(x) = \lim_{x \to 1} (7x-2) = 5$

$\text{R.H.L.} = \lim_{x \to 1^+} f(x) = \lim_{x \to 1} Kx^2 = K$

Since the function is continuous at $x=1$ so,

L.H.L. = R.H.L

i.e. $K = 5$ ※.

Problem: Find a value of the constant K s.t. it will make the following function continuous at $x=2$:

$$f(x) = \begin{cases} Kx^2 & , x \le 2 \\ 2x+K & , x > 2 \end{cases}$$

Sol^n: Do Yourself !!!

A H M Mahbubur Rahman, PhD
Assistant Professor
MNS Dept.
BRACU

Scanned with CamScanner

Extra problems:

1. For what value of a is $f(x) = \begin{cases} x^2 - 1, & x < 3 \\ 2ax, & x \geq 3 \end{cases}$ continuous at every x ?

2. For what values of a is $f(x) = \begin{cases} a^2x - 2a, & x \geq 2 \\ 12, & x < 2 \end{cases}$ continuous at every x ?

3. For what values of b is $g(x) = \begin{cases} \frac{x-b}{b+1}, & x < 0 \\ x^2 + b, & x \geq 0 \end{cases}$ continuous at every x ?

4. For what values of a and b is $f(x) = \begin{cases} -2, & x \leq -1 \\ ax - b, & -1 < x < 1 \\ 3, & x \geq 1 \end{cases}$ continuous at every x ?

Scanned with CamScanner

Lecture 03

Recall the Rates of Change

We need to look at the rate of change problem.

Here we are going to consider a function $f(x)$ that represents some quantity that varies as x varies.

What we want to do here is determine just how fast $f(x)$ is changing at some point from $x = x_1$ to $x = x_0$ over the interval $[x_0, x_1]$. This is called average rate of change of $f(x)$ at $x = x_0$ over the interval $[x_0, x_1]$ and we write

$$r_{ave} = \frac{f(x_1) - f(x_0)}{x_1 - x_0} = \frac{\text{Change in } f(x)}{\text{Change in } x}$$

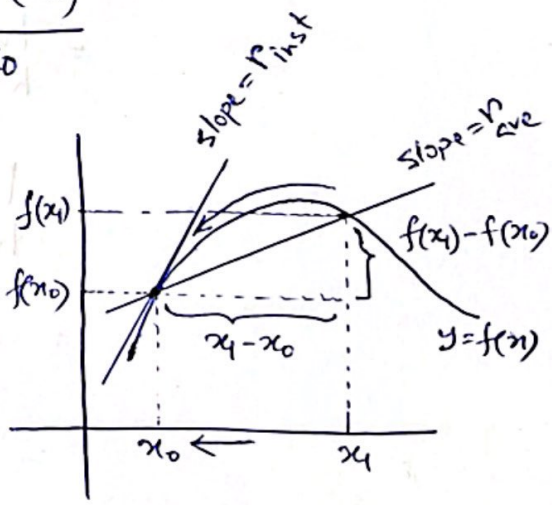
To estimate the instantaneous rate of change at $x = x_0$, we need to do is to choose values of x getting closer and closer to $x = x_0$.

For instatataneous rate of change we can write,

$$r_{insl} = \lim_{x_1 \to x_0} \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

let $h = x_1 - x_0$, so

$$r_{inst} = \lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h}$$



Scanned with CamScanner

If the previous limit exists i.e. $\lim_{h \to 0} \frac{f(x_0+h) - f(x_0)}{h}$ exists, then it can be interpreted as the slope of the tangent line to the curve $y = f(x)$ at $x = x_0$.

This limit is so important that it has a special notation:

$$f'(x_0) = \lim_{h \to 0} \frac{f(x_0+h) - f(x_0)}{h}$$

<u>Definition of derivative</u>

The function f' defined by the formula

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

is called the derivative of f with respect to x. $\square$

<u>Example</u>: Find the derivative with respect to x of $f(x) = x^2$.

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \to 0} \frac{(x+h)^2 - x^2}{h}$$

$$= \lim_{h \to 0} \frac{\cancel{x^2} + 2xh + h^2 - \cancel{x^2}}{h}$$

$$= \lim_{h \to 0} \frac{h(2x+h)}{h} \quad , h \neq 0$$

$$= 2x$$

$\square$

Scanned with CamScanner

Problem: Show that

$$f(x) = \begin{cases} x^2+1 & , x \le 1 \\ 2x & , x > 1 \end{cases}$$

is continuous and differentiable at $x=1$.

Solⁿ. If the function is differentiable at $x=1$, then it must be continuous at $x=1$.

$f(x) = x^2+1$

$f(1) = 1^2+1 = 2$

$f(1+h) = (1+h)^2+1$
$= 1+2h+h^2+1$
$= 2+2h+h^2$

Now,

$$f'_-(x) = \lim_{h\to 0^-} \frac{f(x+h)-f(x)}{h}$$

$$f'_-(1) = \lim_{h\to 0} \frac{f(1+h)-f(1)}{h}$$

$$= \lim_{h\to 0} \frac{\cancel{2}+2h+h^2-\cancel{2}}{h}$$

$$= \lim_{h\to 0} \frac{h(2+h)}{h} = 2$$

$f(x) = 2x$

$f(1) = 2$

$f(1+h) = 2(1+h)$
$= 2+2h$

and,

$$f'_+(x) = \lim_{h\to 0^+} \frac{f(x+h)-f(x)}{h}$$

$$f'_+(1) = \lim_{h\to 0} \frac{f(1+h)-f(1)}{h}$$

$$\lim_{h\to 0} \frac{\cancel{2}+2h-\cancel{2}}{h}$$

$$\lim_{h\to 0} \frac{2h}{h} = 2$$

$$f'_-(1) = f'_+(1)$$

Therefore $f(x)$ is differentiable at $x=1$.

Scanned with CamScanner

Problem: Show that $f(x) = \begin{cases} x^2+2 & x \leq 1 \\ x+2 & x > 1 \end{cases}$

is continuous but not differentiable at $x = 1$.

$\boxed{\text{Do yourself !!!}}$

Scanned with CamScanner

Problem: Find the derivative with respect to x of $f(x) = x^3 - x$.

Sol^n:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \to 0} \frac{(x+h)^3 - (x+h) - [x^3 - x]}{h}$$

$$= \lim_{h \to 0} \frac{\cancel{x^3} + 3x^2h + 3xh^2 + h^3 - \cancel{x} - h - \cancel{x^3} + \cancel{x}}{h}$$

$$= \lim_{h \to 0} \frac{3x^2h + 3xh^2 + h^3 - h}{h}$$

$$= \lim_{h \to 0} \frac{h(3x^2 + 3xh + h^2 - 1)}{h}$$

$$= \lim_{h \to 0} (3x^2 + 3xh + h^2 - 1)$$

$$= 3x^2 - 1$$

$\square$

EXTRA PROBLEMS

PAGE: 154

PROBLEMS: 47 – 50.

Scanned with CamScanner